

A.Y. 2025/2026

Presented by

Alice Buccoleri s358621

Daniele Bongiovanni s358622

AROL CUSTOMER PLATFORM

Conversational Platform for industrial fleet management

Politecnico di Torino - System and Device Programming

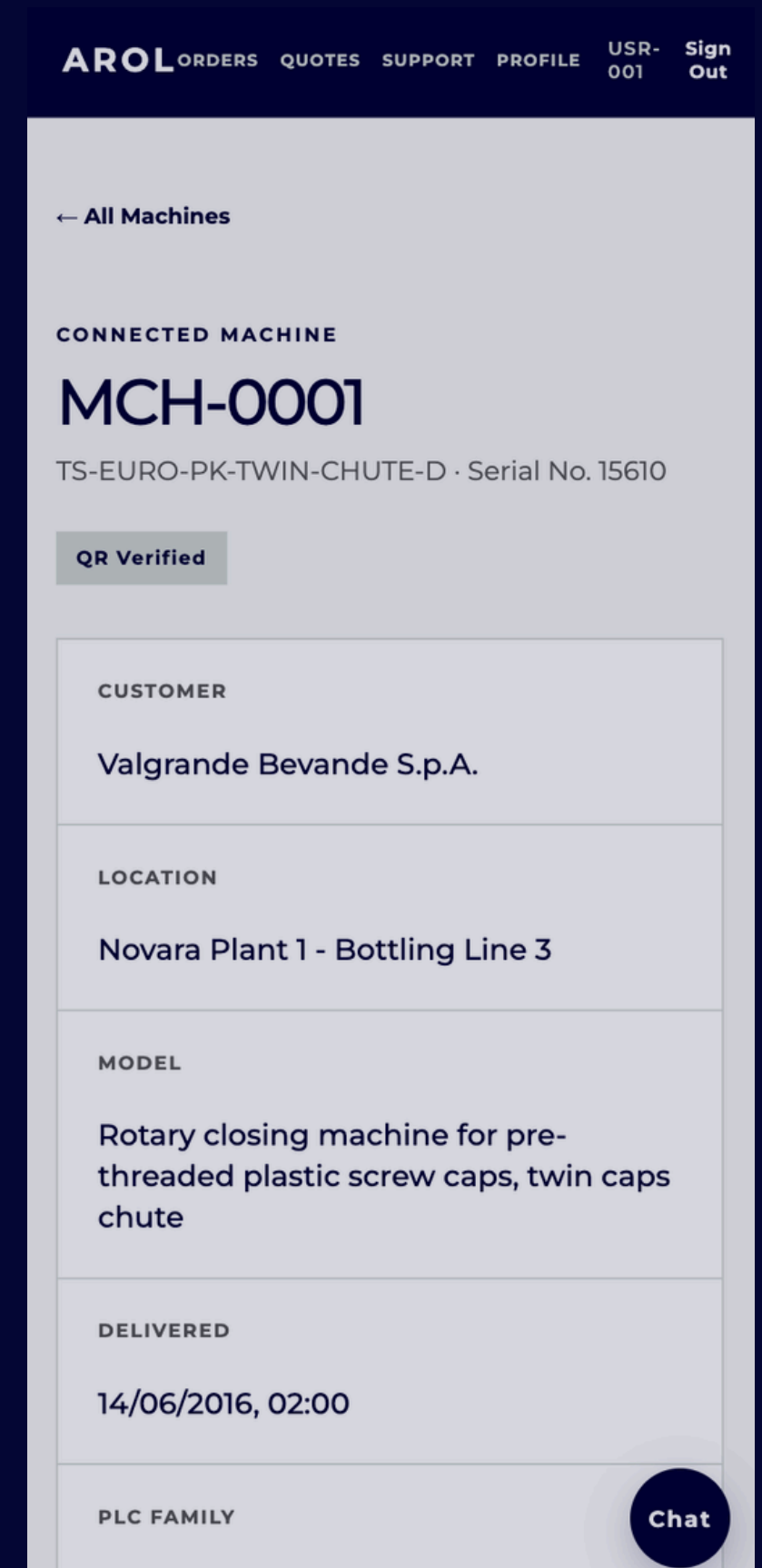
PROJECT AIM

A **single authenticated platform** for AROL customer companies

- Fleet, operational, service and commercial information
- Machine-specific technical manuals
- Direct consultation and conversational access
- Organisation and role-based visibility

Platform Features and Frontend Experience

- Fleet overview and machine-specific pages
- QR-code lookup for direct machine context
- Operational data, tickets, manuals, quotations and orders
- AROL Assistant with structured answers and citations
- Responsive React interface inspired by the AROL visual style



ACCESS CONTROL AND PRIVACY

Backend enforcement

The LLM cannot access PDFs, databases, SQL queries or internal functions.

- JWT-based authentication and trusted AuthContext
- Organisation-level tenant isolation
- Role and machine-context checks
- Parameterised SQL queries
- Local manual processing and embeddings

User visibility profiles

User role	Accessible information
Full visibility user	operational, technical and commercial data
Technician user	machines, telemetry, alarms, tickets and manuals
Commercial user	machines, manuals, quotations and orders

YOUR EQUIPMENT

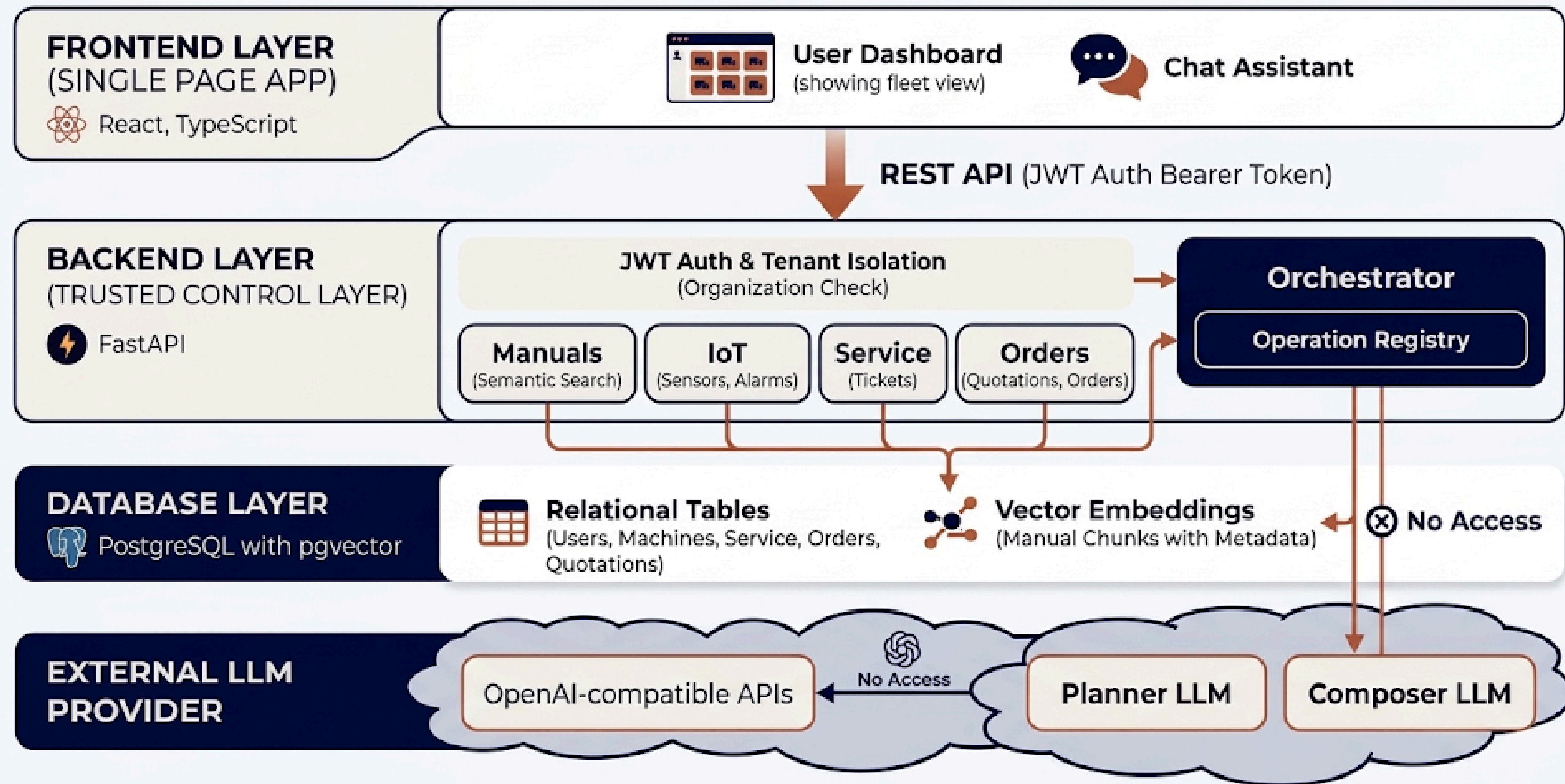
Connected Machines

Select a machine to view its technical documentation and telemetry, or scan a QR code to link a new one.

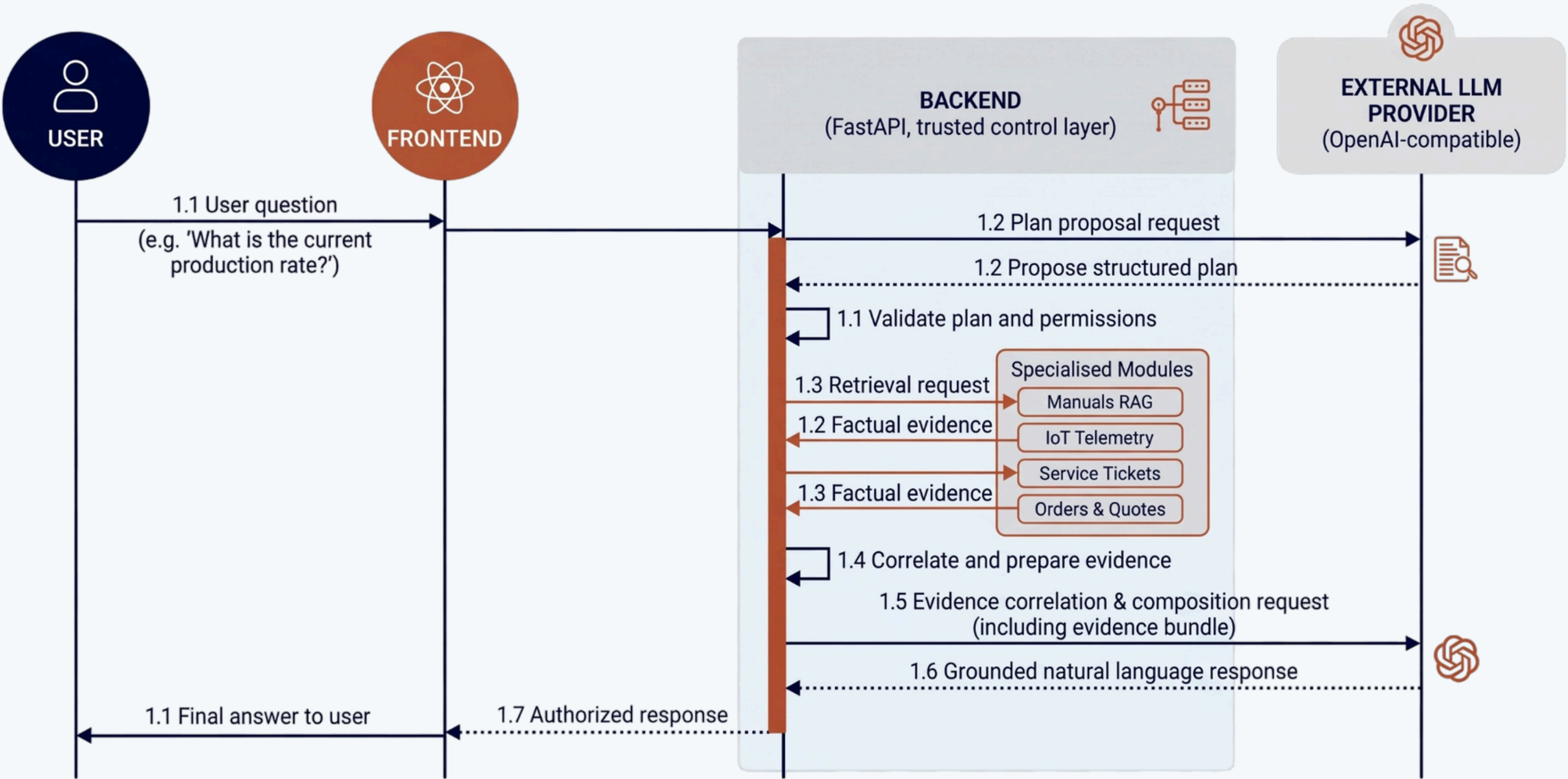
↩ Scan QR code

MCH-0001 → TS-EURO-PK-TWIN-CHUTE-D Novara Plant 1 - Bottling Line 3 Serial 15610	MCH-0002 → TSI-EURO-VA-IES-LIGHT Novara Plant 1 - Bottling Line 5 Serial 17203	MCH-0003 → TSI-EURO-VA-IES-LIGHT Alessandria Plant 2 - Bottling Line 1 Serial 17579
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SYSTEM ARCHITECTURE



CONVERSATIONAL REQUEST FLOW

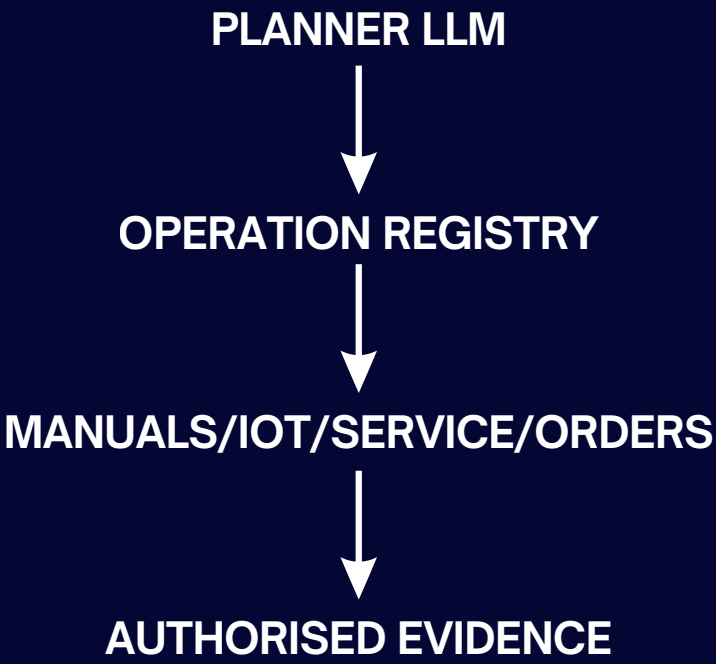


The backend executes operations. The LLM never executes tools directly.

OPERATION REGISTRY

- 1.Receives the Planner LLM request
- 2.Verifies that the operation is registered
- 3.Validates typed parameters with Pydantic
- 4.Requires trusted machine context when needed
- 5.Dispatches the authorised handler
- 6.Returns a standardised AgentResult

Unknown operations or invalid parameters are rejected before data retrieval.



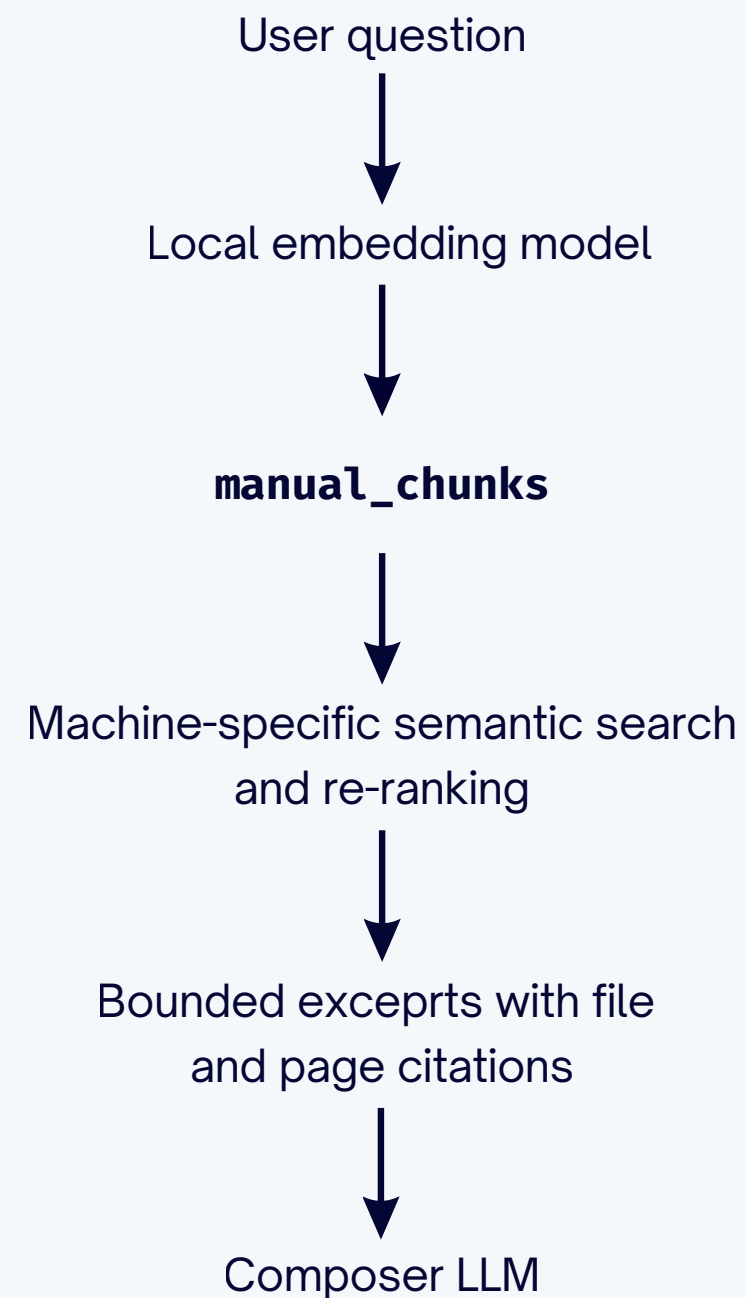
MCP concept	Our implementation
Tool catalogue	<code>planner_catalog()</code>
Tool schema	Pydantic models
Tool invocation	<code>OperationRegistry.execute()</code>
Tool result	<code>AgentResult</code>

LOCAL RETRIEVAL WORKFLOWS

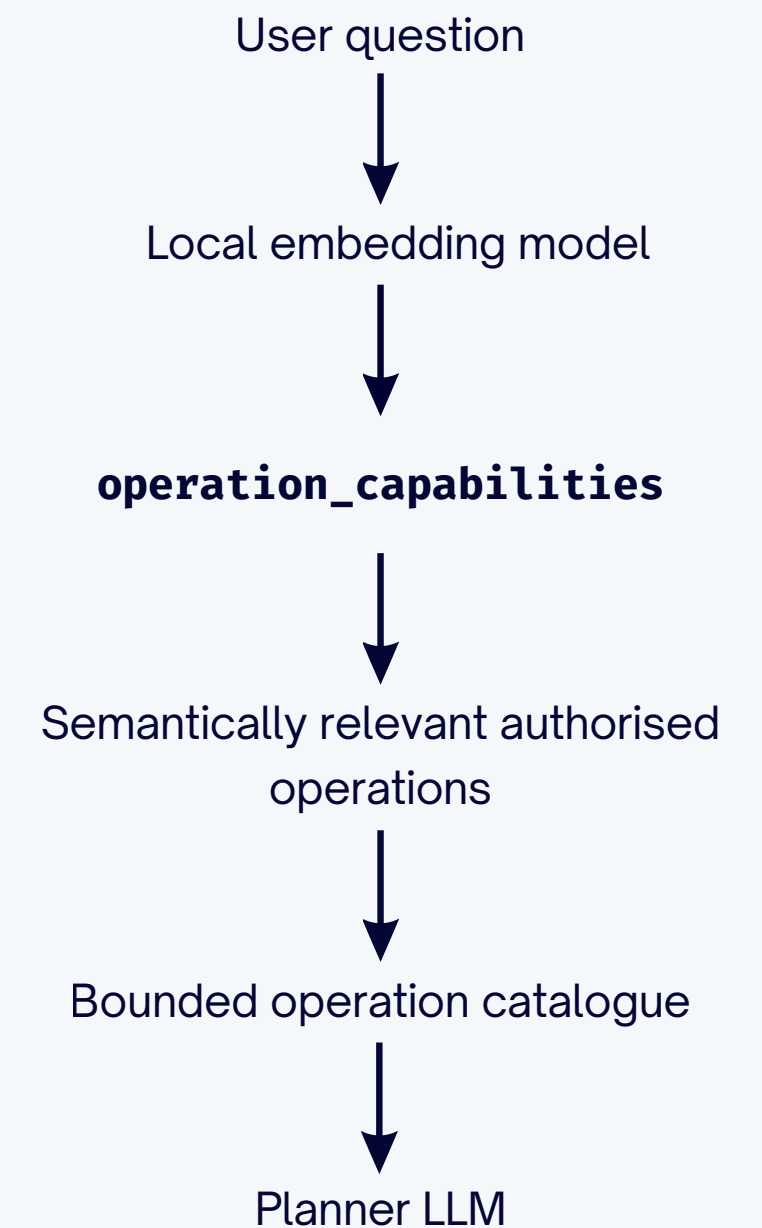
Manual RAG retrieves factual technical evidence.
Capability retrieval helps the Planner choose authorised operations.

Both indexes are local and use the same embedding model.

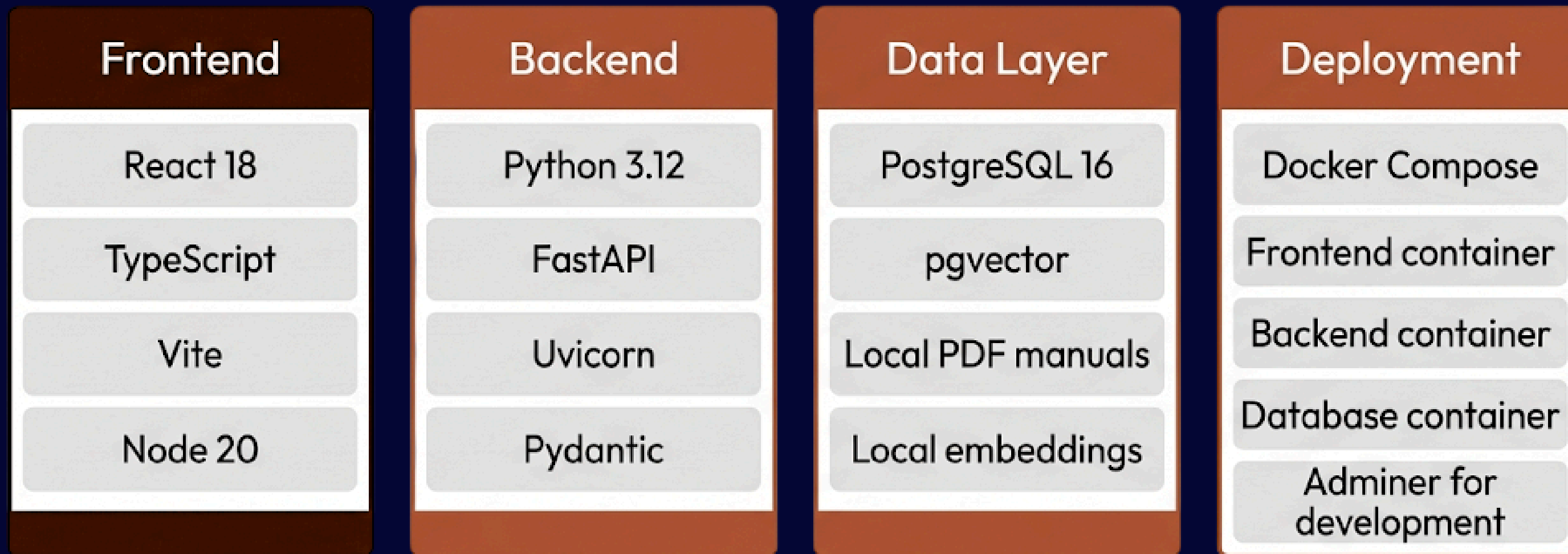
Manual RAG



Planner capability retrieval



DEPLOYMENT AND TECHNOLOGY STACK



Configuration is provided through a local `.env` file.
Manuals, embeddings and database access remain inside the local environment.

Testing and Evaluation

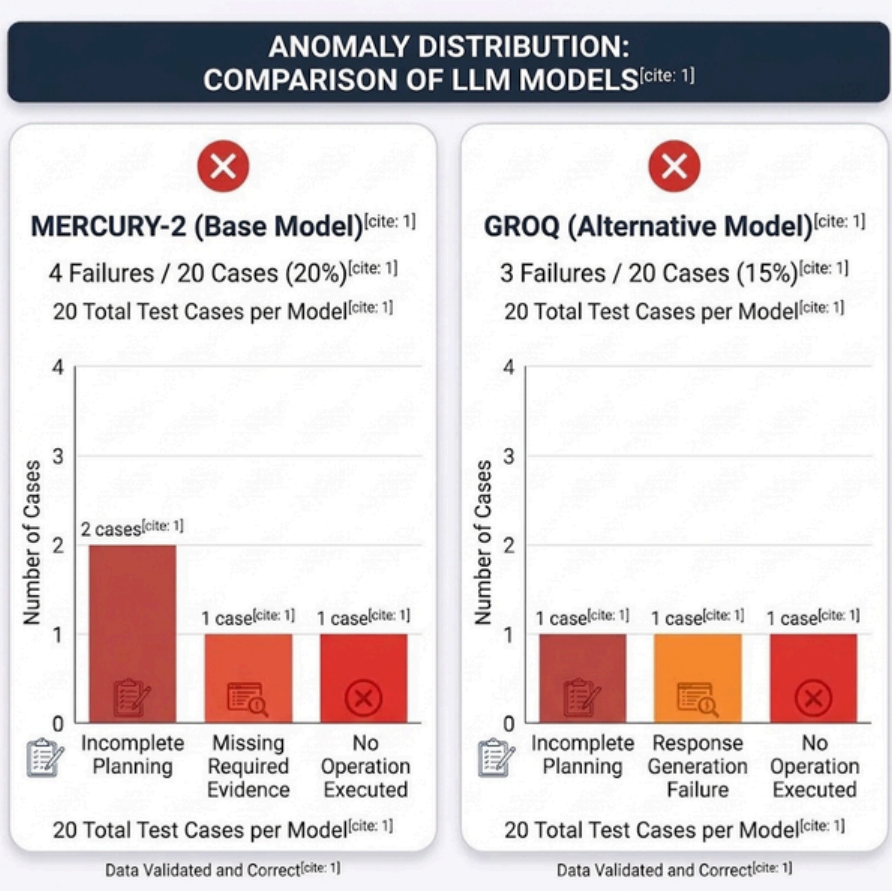
Automated Validation

- Backend, integration and frontend **test suites passed**
- Authentication, tenant isolation and role checks verified
- API contracts, orchestration and local retrieval tested

Metric	Result
 Evaluation cases	16
 File Recall@K	1.00
 Mean Reciprocal Rank	1.00
 Mean keyword coverage	0.714
 Page Recall@K	0.75
 Page Recall@K within tolerance	0.94

“The relevant manual file was consistently retrieved flawlessly. Page-level retrieval precision is now highly robust, with the exact or an adjacent page retrieved in **94%** of cases.”

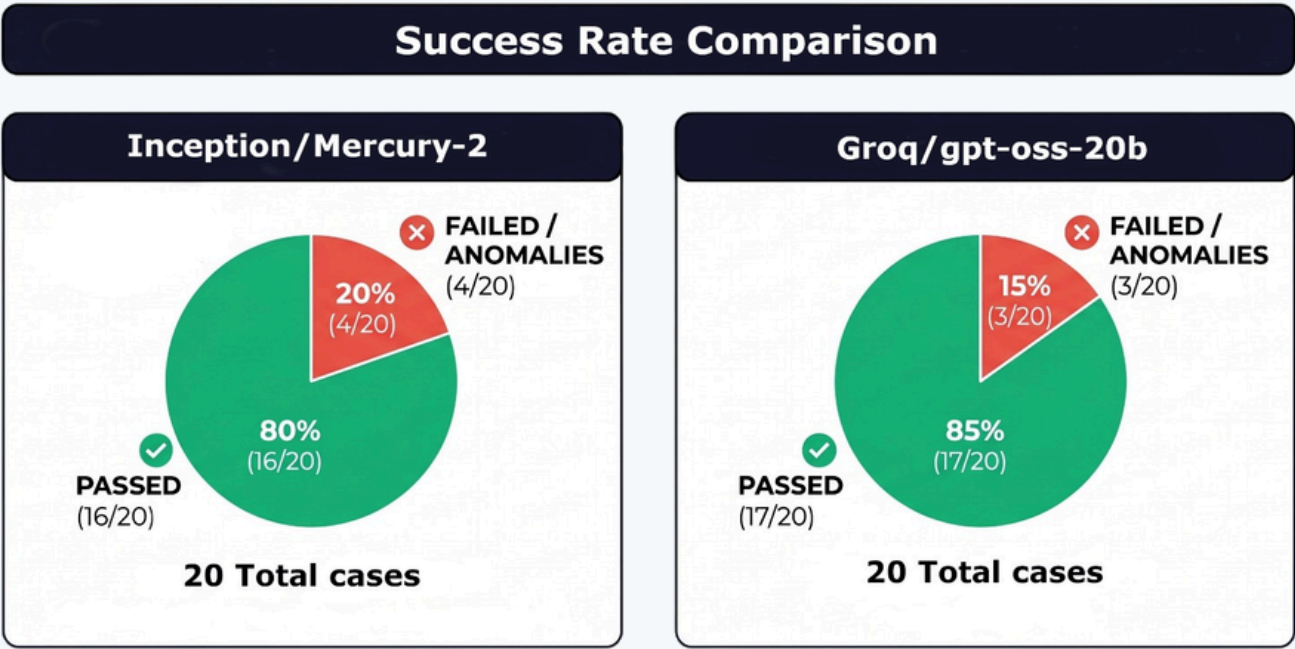
Testing and Evaluation



Response Latency Comparison (Response Latency Benchmark)

Metric	Mercury-2 (Base Model)	Groq (openai/gpt-oss-20b)
Average Latency	3.41 s	27.69 s
Fastest Time (Min)	1.48 s	7.99 s
Slowest Time (Max)	10.64 s	48.30 s

Note for report: Mercury-2 is approximately 8 times faster on average, fundamental for an interactive chatbot experience.



CONCLUSION

Next steps...

Future work will focus on evaluating **stronger reasoning models** and **improving the orchestration prompts, contextual evidence representation**, and review of **multi-source plans**, while preserving the backend as the deterministic security and authorisation boundary.

Project Outcomes

- A unified platform for fleet, service, commercial records and machine-specific manuals
- Organisation and role-based access control enforced by the backend
- Local RAG with traceable manual citations
- A custom orchestrator that controls LLM access through the Operation Registry

Current Limitations

- The Planner may omit a required operation
- A response may lack required evidence
- Some workflows may end without an executed operation
- Page-level manual retrieval can be improved

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THANK YOU

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